SECURELINK VPN LAB

Research Project | Secure VPN for Remote Access

Project Index and Traceability

Establishing a Secure VPN for Remote Access

Submission pack

Organisation	Research project (lab build)
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Version	1.0 (final)
Basis	Lab build on VMware; evidence IDs [E01]-[E38] refer to the evidence screenshot set
Status	Lab documentation. Not certified and not presented as production-ready.

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1 Documents in this pack

No.	Document	File
00	Index and Traceability	00_Index_Traceability.docx
D1	VPN Architecture Design Document	D1_VPN_Architecture_Design.docx
D2	Encryption & Authentication Plan	D2_Encryption_Authentication_Plan.docx
D3	VPN Client Configuration Guide	D3_Client_Configuration_Guide.docx
D3A	Windows OpenVPN Test Report (appendix to D3)	D3A_Windows_OpenVPN_Test_Report.docx
D4	Monitoring & Logging Plan	D4_Monitoring_Logging_Plan.docx
D5	Performance & Scalability Report	D5_Performance_Scalability_Report.docx
D6	Policy & Compliance Documentation	D6_Policy_Compliance_Documentation.docx

The brief names six deliverables (D1 to D6); the index is an additional cover document. Evidence: 38 screenshots, E01 to E38.

2 Traceability matrix

Brief task	Deliverable	Evidence	Status
1 Architecture: topology	D1	E01, E05, E06, E10, E16	Implemented
1 Architecture: technology selection	D1	E01, E05	Implemented; alternatives assessed on paper
2 Encryption: AES-256, end-to-end	D2	E01, E05, E10	Implemented
3 MFA	D2, D3	E05, E07, E08, E09	Implemented
3 Access control	D2, D6	E11, E12	Partial (network-based)
4 SHA-256 integrity	D2	E10, E17, E18	Implemented
4 Certificates / non-repudiation	D2	E02, E03, E26, E23-E25	Partial (one certificate, single-tier CA)
5 Clients: Linux	D3	E04, E05, E13, E14	Implemented
5 Clients: Windows, macOS	D3, D3A	E29-E38	Windows tested; macOS not tested
5 Compliance checks	D3	E13, E14, E15	Partial (client-side, bypassable)
6 Monitoring and logs	D4	E09, E11, E17, E19, E27, E28	Partial (no alerting, dashboards)
7 Load balancing	D5	E16, E21, E22	Partial (client-chosen, one gateway)
7 Scalability	D5	E15, E20	Not implemented (recommendations)

8 Policies	D6	E23-E25	Implemented as documents
8 GDPR and HIPAA	D6	E05, E11, E28	Partial (alignment mapping only)

3 Limitations summary

- Lab on one shared Windows 10 host; LANs are simulated with dummy interfaces.
- Single-tier CA; one client certificate and one account; no per-user accountability.
- Compliance script is client-side, Linux-only and bypassable.
- Syslog over plain TCP 514; no dashboards, alerting or retention.
- Load balancing is two instances on one gateway; performance test used one client.
- macOS client, two-tier PKI, TLS syslog and multi-gateway HA were not done.
- ELK, Nagios, Zabbix, FreeRADIUS, WireGuard and SoftEther were not deployed.
- GDPR and HIPAA content is an alignment mapping, not legal advice.

4 Authorship statement

I designed, built and tested this lab myself (Kshitij Rajesh Randhire): the configurations, test runs and all 38 evidence screenshots come from my own lab work. I used Claude, an AI assistant made by Anthropic, only to help draft and edit the wording of the documents from my evidence and notes. The results, limitations and conclusions stated here are those shown by my lab evidence, and I am responsible for the content.